

Introduction

One of the most significant obstacles to thriving in our changing world is understanding and anticipating the global atmospheric environment, specifically the impacts on climate and human health. To reduce the risk of disease and save human lives, accurate global estimates of air pollution, such as ozone, are necessary to assess the global public health burden of disease linked to air pollution exposure. This work aims to provide new scientific insights into the drivers of surface ozone trends at scales relevant to assessing human health impacts. To achieve the goal, as a first step, we employed machine learning (ML) based feature permutation importance analysis to derive insights into the sub-scale drivers of the residual bias between surface observations and chemical transport model predictions.

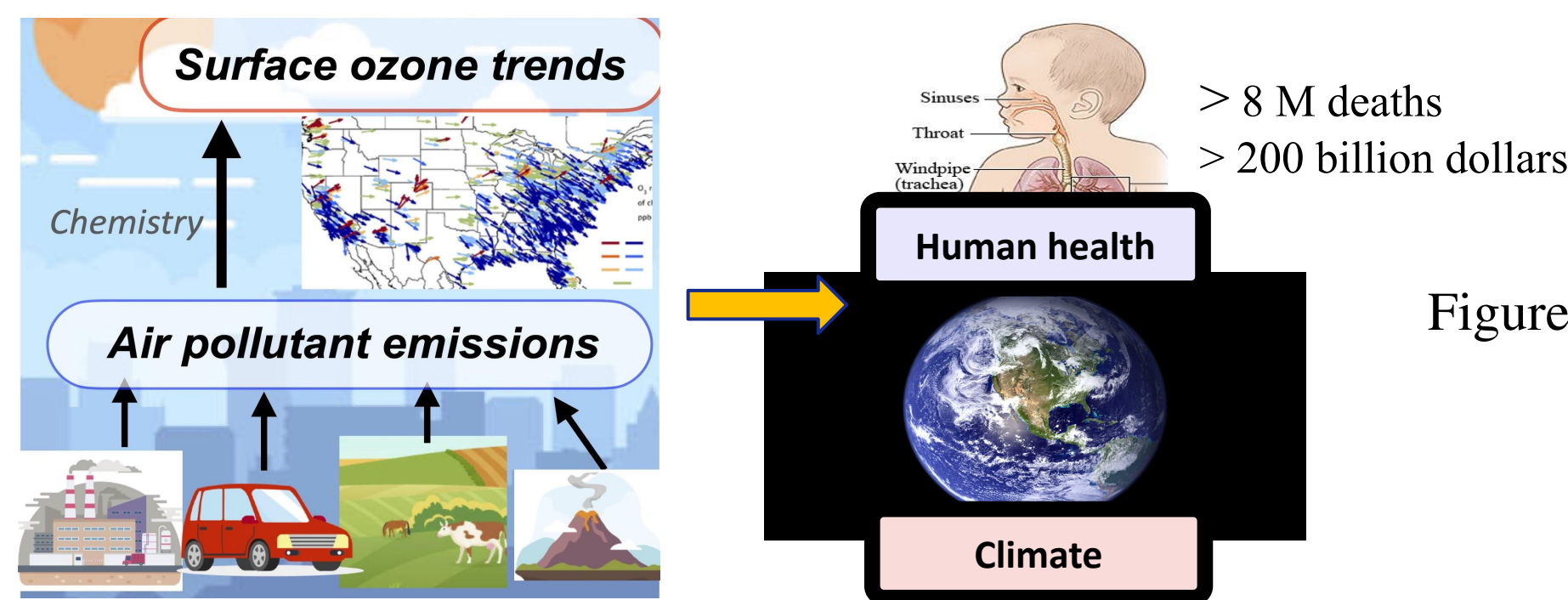


Figure 1. Diagram of the effects of air pollution

Objective: Understanding the sub-scale drivers of the surface ozone trends to decrease the residual bias between surface observations and chemical transport model predictions

Solution: Using ML permutation importance analysis to understand sub-scale drivers of the of the residual bias between surface observations and chemical transport model predictions

1. Data Sets

To build an ML model and perform feature permutation importance analysis, we used the data sets from:

- Tropospheric Ozone Assessment Report, Phase 2 (TOAR-2) surface observations.
 - TOAR-2 database contains the largest collections of near-surface ozone measurements, mainly covering North America, Europe, and East Asia.
 - A compiled TOAR-2 database, the Air Quality Benchmark (AQ-Bench) (Betancourt et al., 2021), includes metadata and aggregated metrics at 5,577 measurement stations all over the world from the years 2010 to 2014, was used as observation input data.

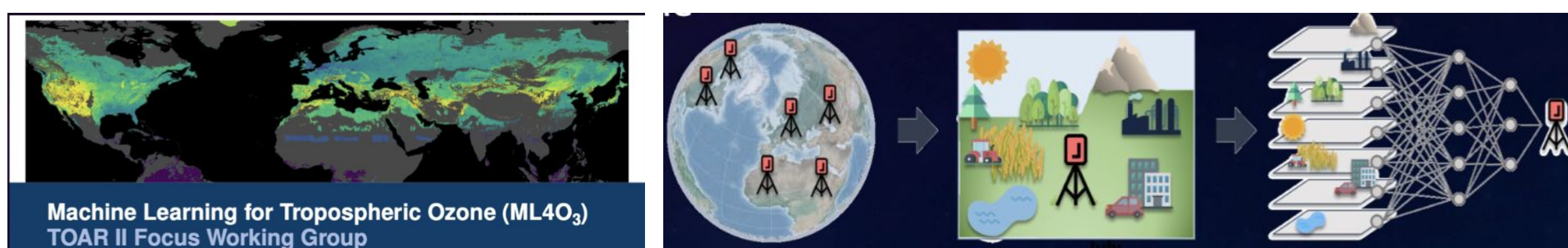
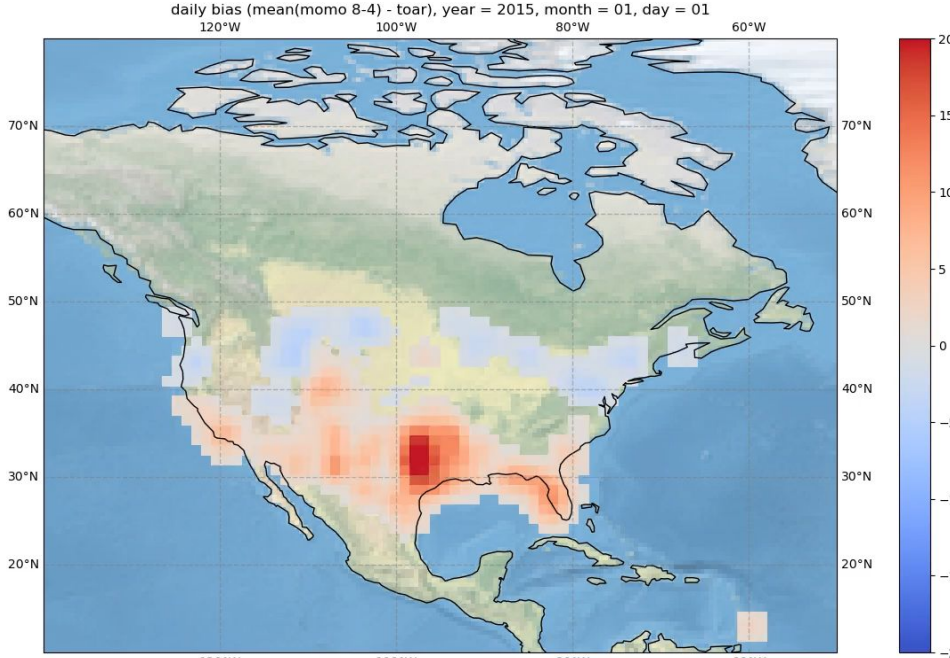


Figure 2. A depiction of the TOAR-2 database, specifically AQ-Bench.

- Multi-model Multi-Constituent Chemical (MOMO-Chem) data assimilation framework
 - The MOMO-Chem framework, developed at JPL (Miyazaki et al., 2020), combines a chemical transport model (CTM) and multiple satellite observations using data assimilation and provides global predictions of ozone and many other atmospheric compositions.

Figure 3. Ozone bias (difference between mda8 and TOAR-2 data) for MOMO-chem over North America, 2015



- High-resolution satellite imagery obtained from Google Earth Engine (GEE).
 - GEE combines a catalog of satellite imagery and geospatial data sets with global analysis capabilities. Specifically, we used the land cover imagery from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) at 500 meters resolution, which would supplement ML predictions by providing detailed spatial patterns relevant to chemical regimes of ozone globally.

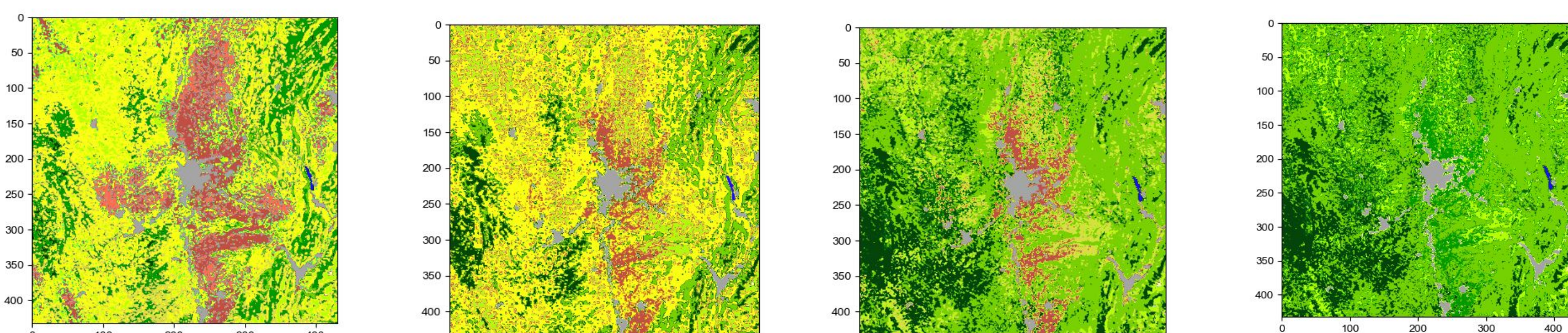


Figure 4. Landcover images that provide high-resolution (500 m) information to be a potential subdriver

2. Methodology

In this work, we are interested in building an ML regression model that maps the input variables to output ozone residual bias.

- Random Forest is an ensemble learning algorithm that uses bootstrap aggregation techniques to train independent decision trees and combine results by averaging the responses of these decision trees.
- In the ML field, variables are often referred to as features.
 - Feature permutation importance is an ML model inspection technique that helps to understand how much an ML model depends on certain features.
 - Feature permutation importance analysis is conducted by computing a decrease in an ML model's performance score when a single feature is randomly shuffled.
 - For example, if randomly shuffling a single feature results in a significant decrease in the model's performance score, then it means this feature has a large impact on the model's predictive power.

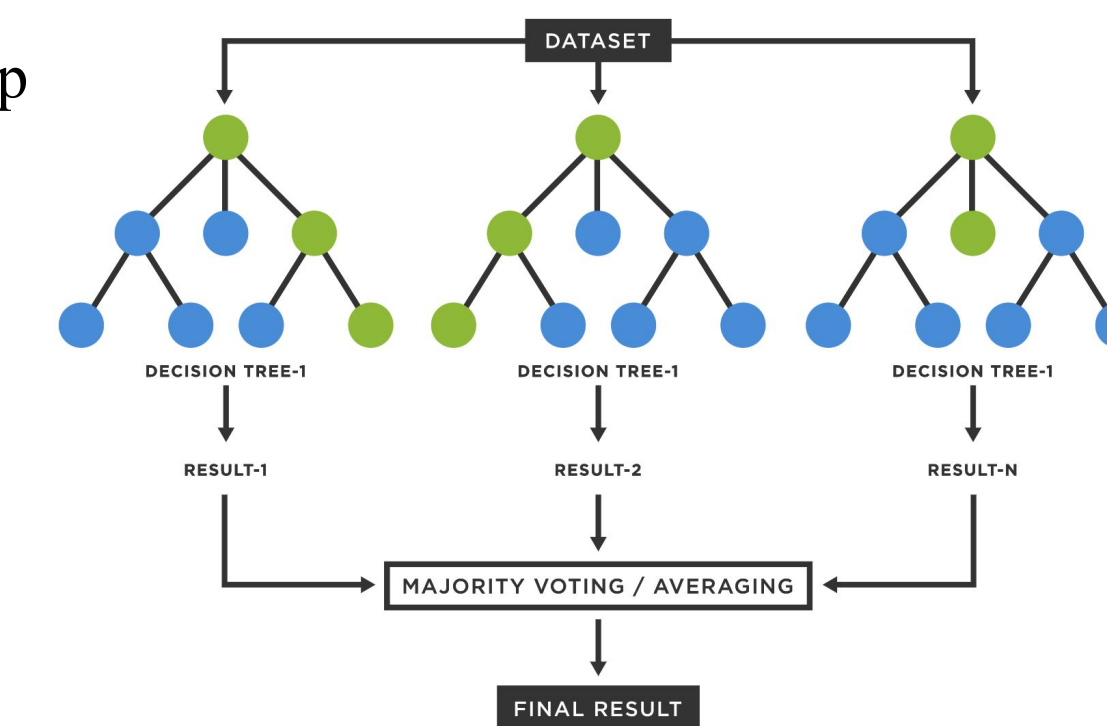


Figure 5. Schematic figure of a Random Forest Model

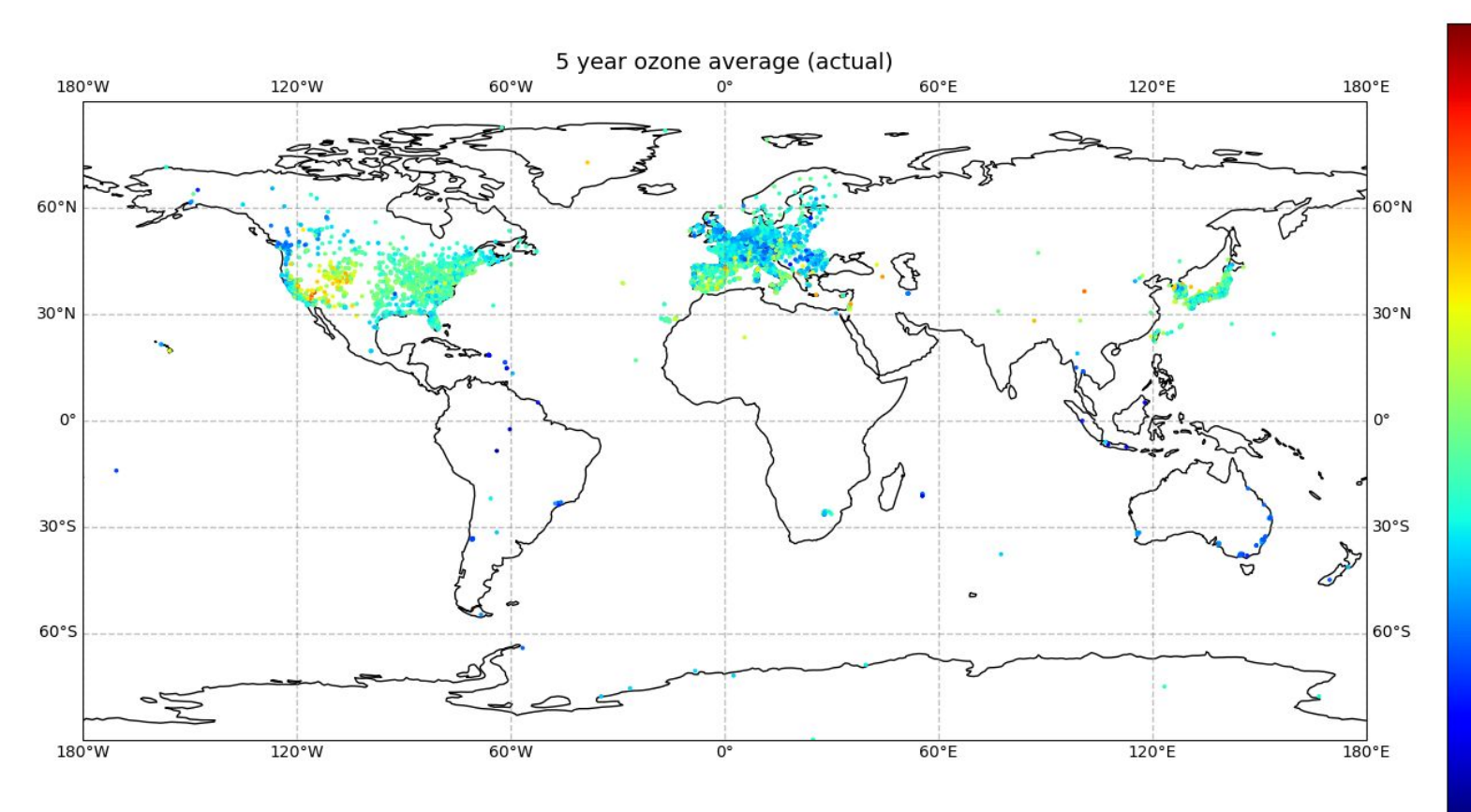


Figure 6. Spatial map of 5-year-average ozone levels (ppb) at each TOAR-2 station from the AQ-Bench dataset

3. Experiments and Results

We conducted two experiments:

- Experiment 1: only the TOAR-2 AQ-Bench multi-parameter data set was used as the input data for the Random Forest model. The target data was the average ozone in the TOAR-2 AQ-Bench data set.
- Experiment 2: four new statistical features (i.e., mean, standard deviation, mode, and the number of different land cover types) derived from the MODIS land cover data set of GEE were used as ML inputs in addition to all the features from TOAR2 AQ-Bench data set.

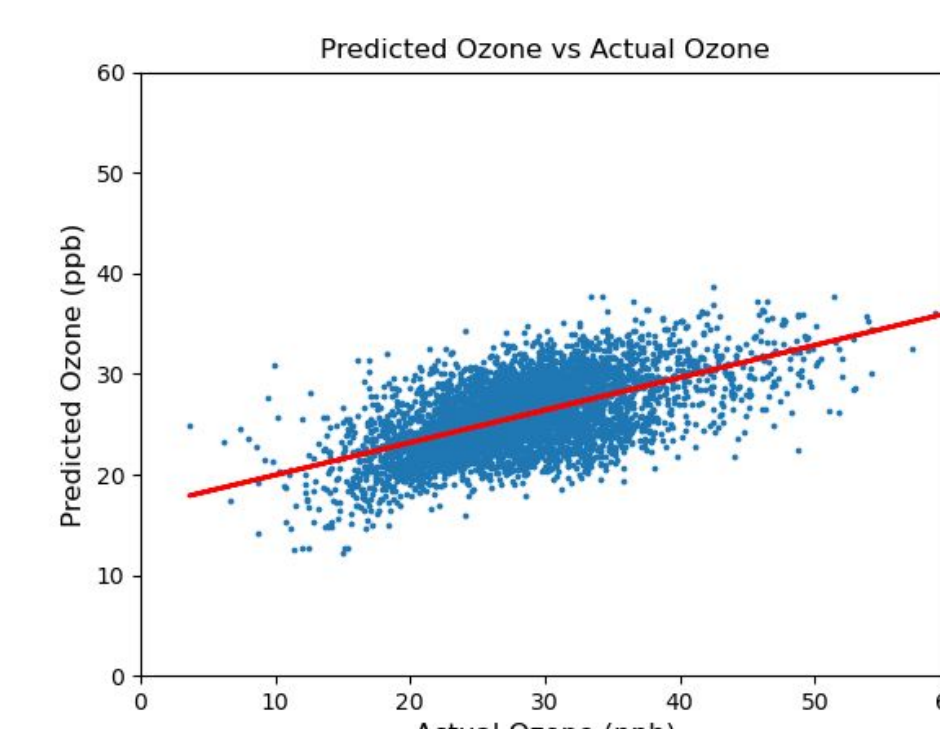


Figure 7. Scatter plot of predicted vs. actual average ozone (ppb) from Experiment 1.

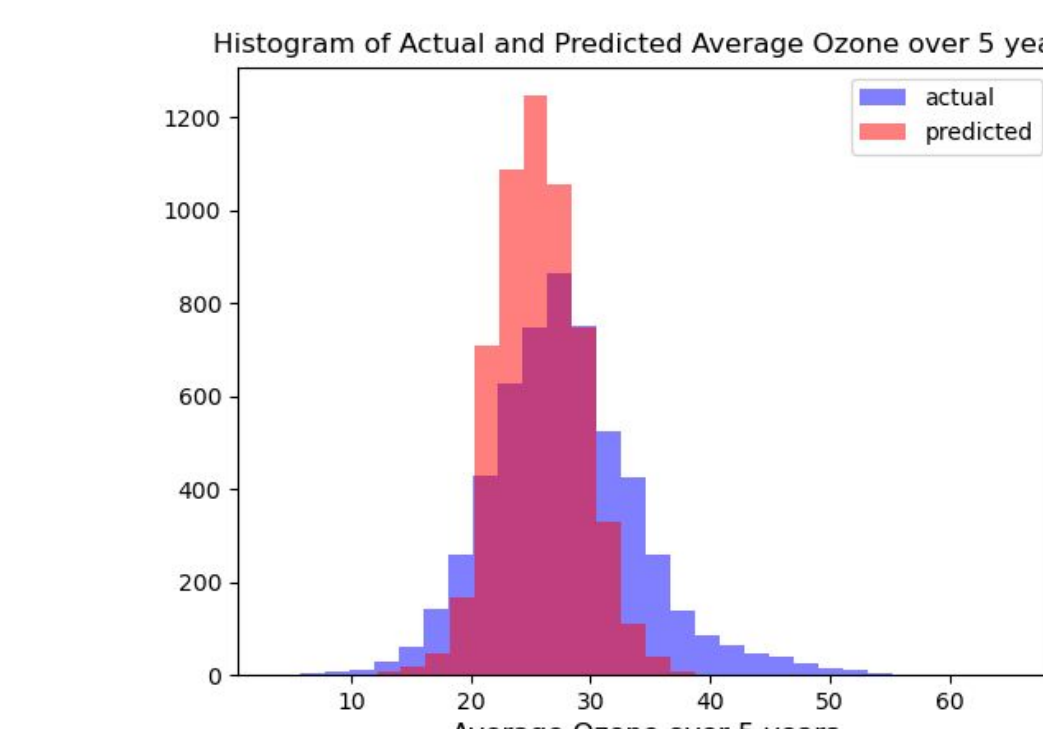


Figure 8. Histogram of actual and predicted average ozone (ppb) from Experiment 1.

The Root Mean Squared Error (RMSE) and Pearson correlation coefficient (Pearson's r) scores are used to measure the correlations of the ground truth and ML predicted ozone values.

	RMSE Score	Pearson's r score
Experiment 1	5.61	0.59
Experiment 2	4.76	0.69

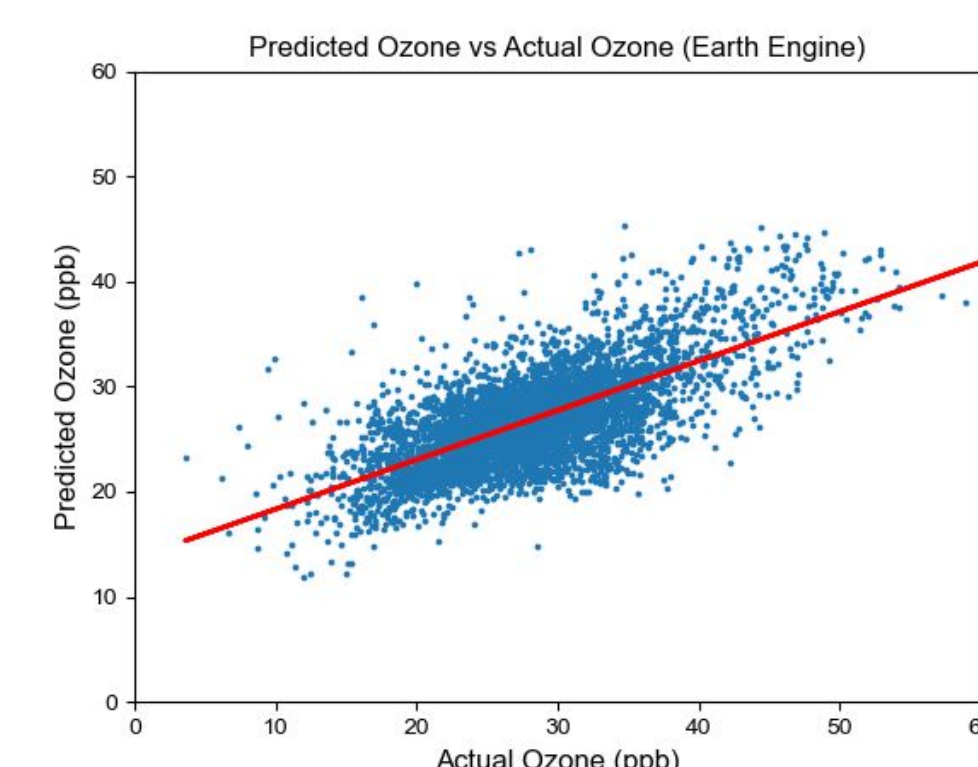


Figure 9. Scatter plot of predicted vs. actual average ozone (ppb) from Experiment 2.

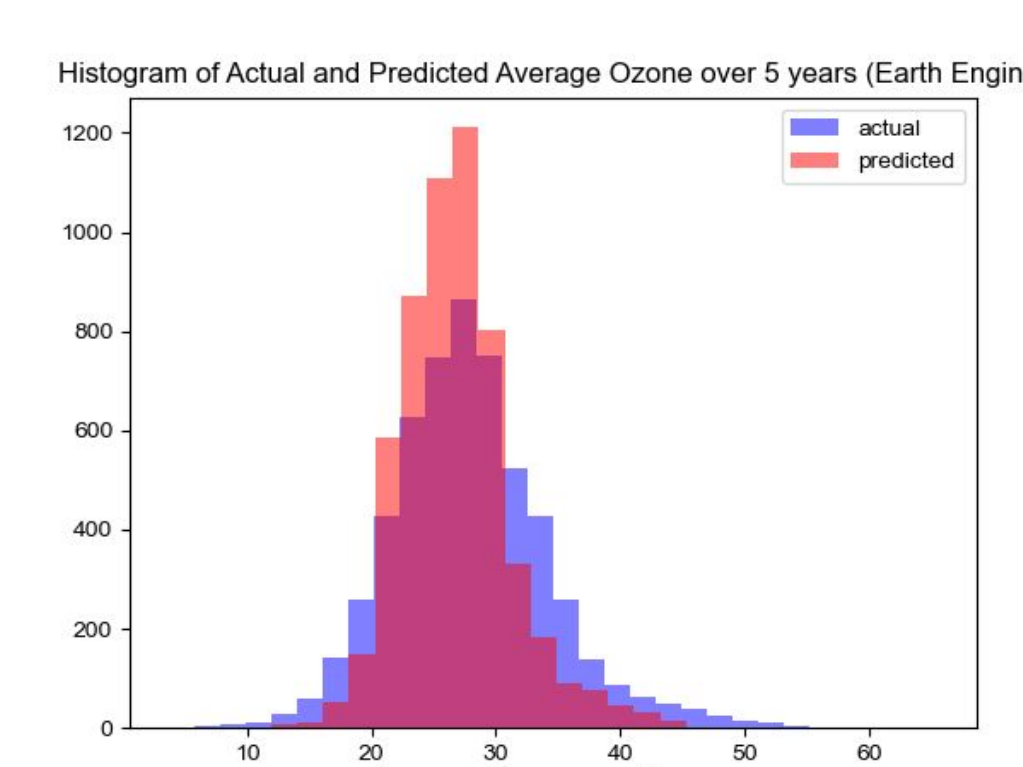


Figure 10. Histogram of actual and predicted average ozone (ppb) from Experiment 1.

3. Experiments and Results (Continued)

- For the AQ-Bench ML experiment:
 - Suggested several highest importance parameters, such as altitudes, nightlight, and NOx emissions to contribute to surface ozone changes based on the feature permutation importance analysis.
 - In this case, cases with high ozone concentrations are not well predicted across the globe, as noted in Figure 8 by the tail of higher actual average ozone values in Experiment 1.
- By adding 4 features from the MODIS land cover data in the ML calculation:
 - In addition to the high importance features noted from Experiment 1, an increase in importance for urban, suburban, and rural indicates their contributions to surface ozone changes.
 - In this case, some polluted cases, with high ozone concentrations, are better predicted across the globe, suggesting that including high-resolution land cover provides important information on the surface ozone chemical regime.

Figure 11. Bar chart of feature and permutation importance from Experiment 1.

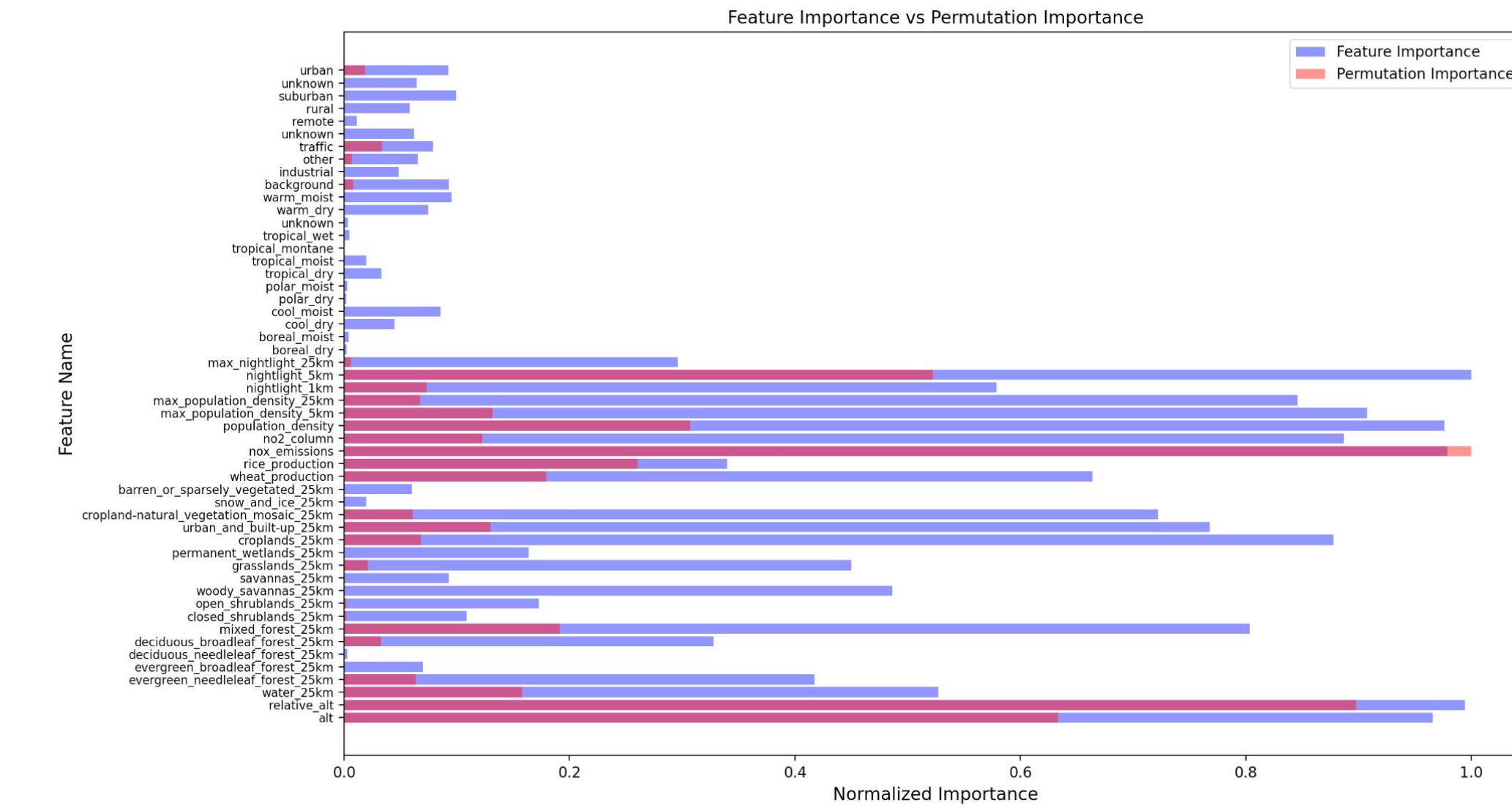
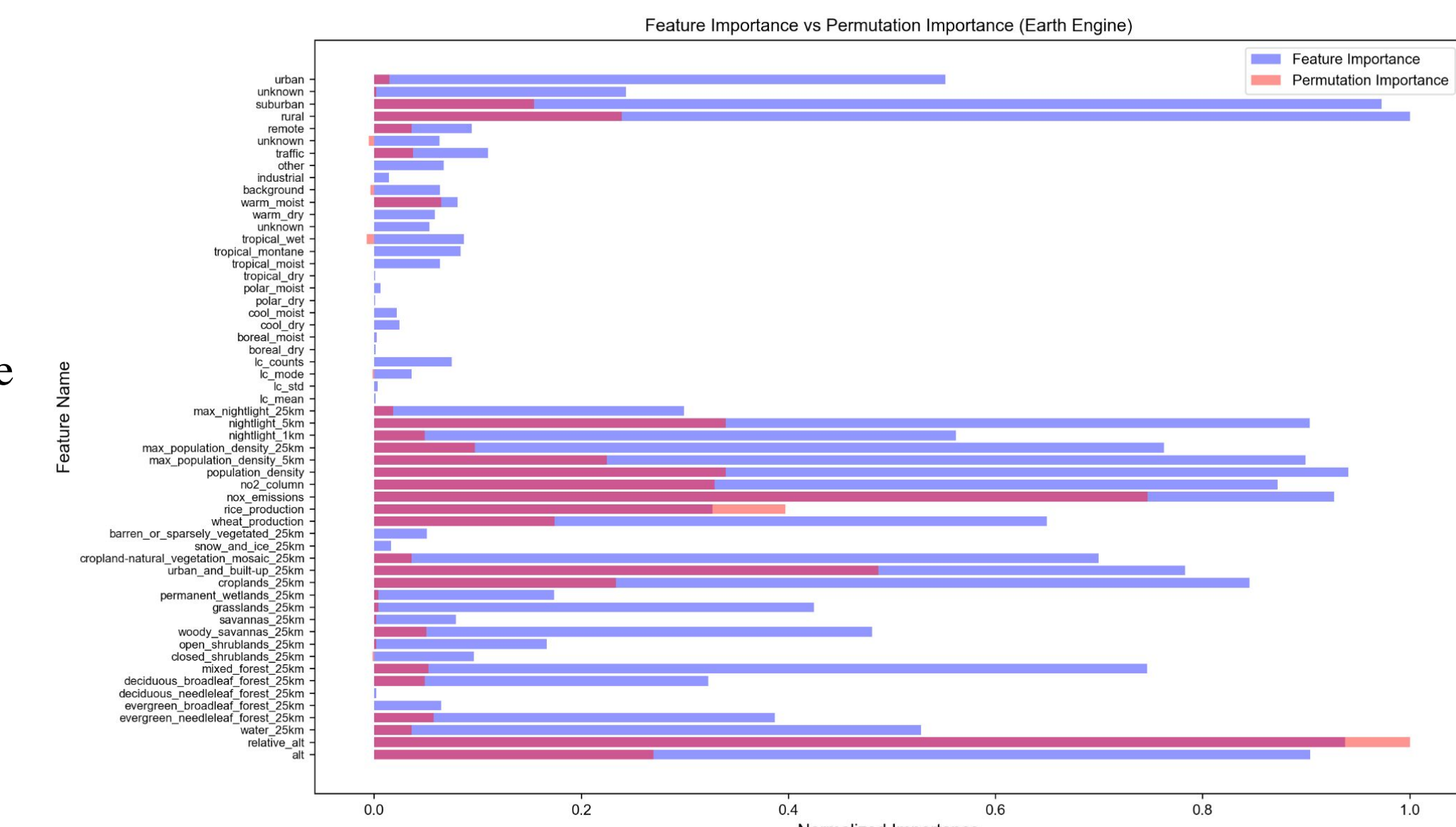


Figure 12. Bar chart of feature and permutation importance from Experiment 2.



4. Conclusions and Future Work

- We introduced a machine learning based feature permutation importance approach to understanding sub-scale drivers of surface ozone.
- We also demonstrated to improve an ML model's predictive power using MODIS land cover data set from GEE.
- In future work, we will:
 - Supply the features identified by the ML models to the MOMO-Chem framework, and verify whether or not these features could help to improve the MOMO-Chem framework.
 - Explore hyperparameter tuning of the Random Forest model; explore other models such as neural networks.
 - Test other high-resolution data in the ML framework through GEE and NASA's data platforms to see whether and how additional data will help fully predict global surface ozone, especially for extremely polluted events to benefit air quality and human health studies.

Acknowledgements

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